

**CRACK INITIATION LIFE CALCULATION FOR
METALLIC MATERIALS**

Part: **4.3 – CRACK INITIATION ANALYSIS**

Chapter: **4320**

Title: **CRACK INITIATION LIFE CALCULATION
FOR METALLIC MATERIALS**

Revision: **1.1**

Date: **18/01/2021**

1 INTRODUCTION

This section includes the description of the general methodology used in Airbus D&S for calculating the crack initiation life of cyclic loaded metallic structures. Other forms of fatigue, such as corrosion fatigue, fretting fatigue, thermal fatigue, fatigue in viscous materials or acoustic fatigue are not covered in this chapter.

The crack initiation is the phase of the total life of the component that covers the nucleation of microscopically small cracks. Thus, the crack initiation life is defined as the total number of cycles before a small crack begins its macroscopic growth.

The crack initiation analysis has three main purposes:

- Demonstrate that no crack initiation will appear in the structure during to the full aircraft DSG.
- Determinate the inspection threshold.
- Residual fatigue life calculation. This subject will be dealt broadly and specifically in future chapters of the Stress Methods Manual.

The crack initiation assessment for a structural element is based on linear cumulative fatigue damage calculation. It is assumed that the failure of the part occurs when the total fatigue damage value exceeds a critical limit $D = 1$. However, when substantiated by test evidences or fleet experience, other values different from the unity could be used.

A description of the general crack initiation analysis methodology is included in the following chapters.

CRACK INITIATION LIFE CALCULATION FOR METALLIC MATERIALS

2 METHOD DESCRIPTION

The crack initiation life calculation methodology deployed in Airbus D&S is based on the use of material endurance data, the calculation of fatigue damage using Palmgren-Miner cumulative damage hypothesis and the use of stress sequence processed by the Rainflow counting method (block spectrum are also used for some legacy programs).

The crack initiation life of a structural component is calculated as:

$$FL = \frac{N_{sp}}{D} = \frac{N_{sp}}{\sum d_i}$$

Where: N_{sp} is the number of flights for which the stress spectrum has been generated.

D is the total damage.

d_i is the damage provided by each cycle of the spectrum.

The damage contribution of each cycle in the spectrum (d_i) is calculated as the inverse of N_i , being N_i the number of cycles up to failure obtained from the appropriate material endurance data (S-N curve, Haigh diagram or ϵ -N curve):

$$d_i = \frac{1}{N_i}$$

The most common approaches to calculate the fatigue damage under cyclic loading are the stress life approach and the strain life approach:

- The stress-life approach is the method most widely used for the crack initiation analysis and the one with the most accessible material data. The stress-life approach is applicable for high cycle fatigue (components failing after more than $5 \cdot 10^3$ - 10^4 cycles) when the material deformation is primarily elastic. For lower crack initiation life the local plasticity is important and the strain-life approach shall be used.
- The strain-life method is better than the stress life approach to account for high stress ranges because it considers local plasticity in the material. This method is applicable to low cycle fatigue (the failure appears before 10^4 cycles), where the material deformation is combination of elastic and plastic strains.

CRACK INITIATION LIFE CALCULATION FOR METALLIC MATERIALS

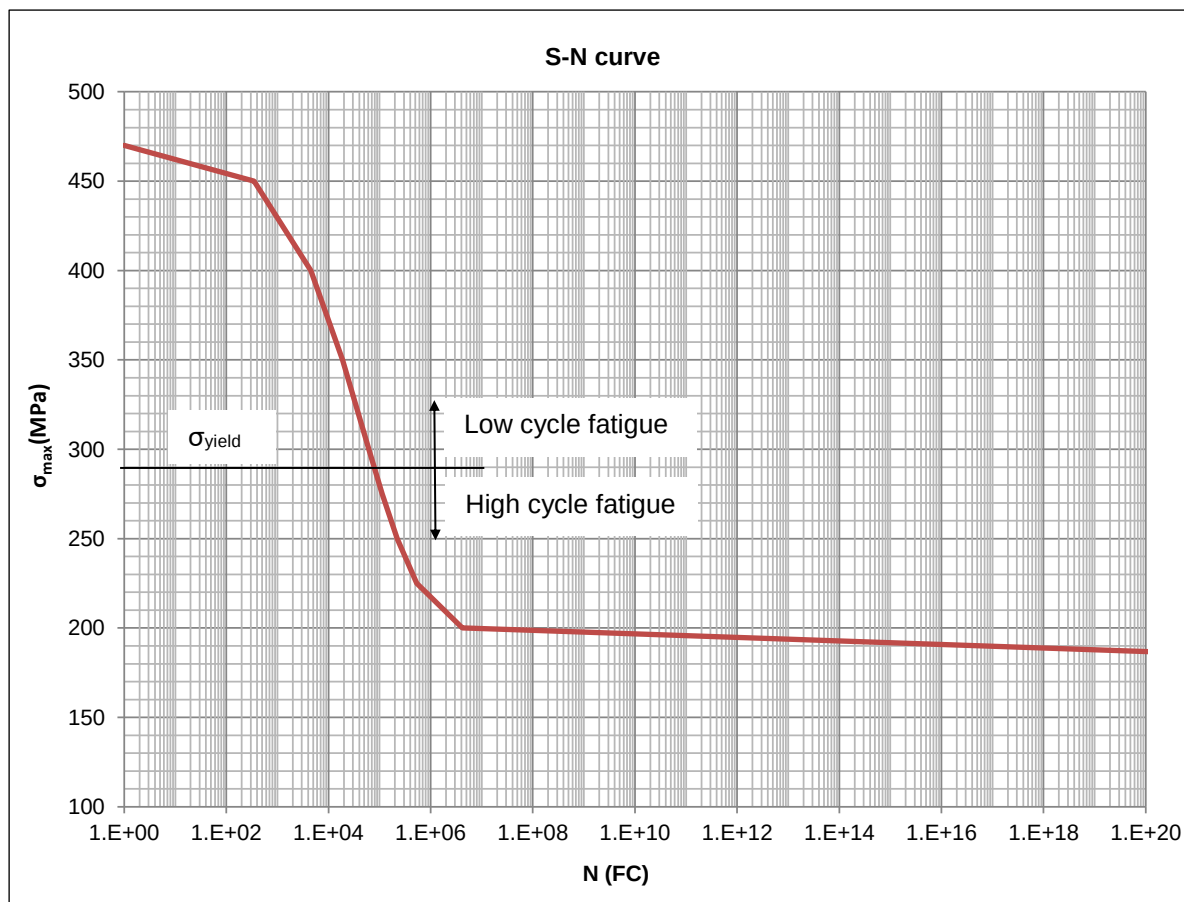
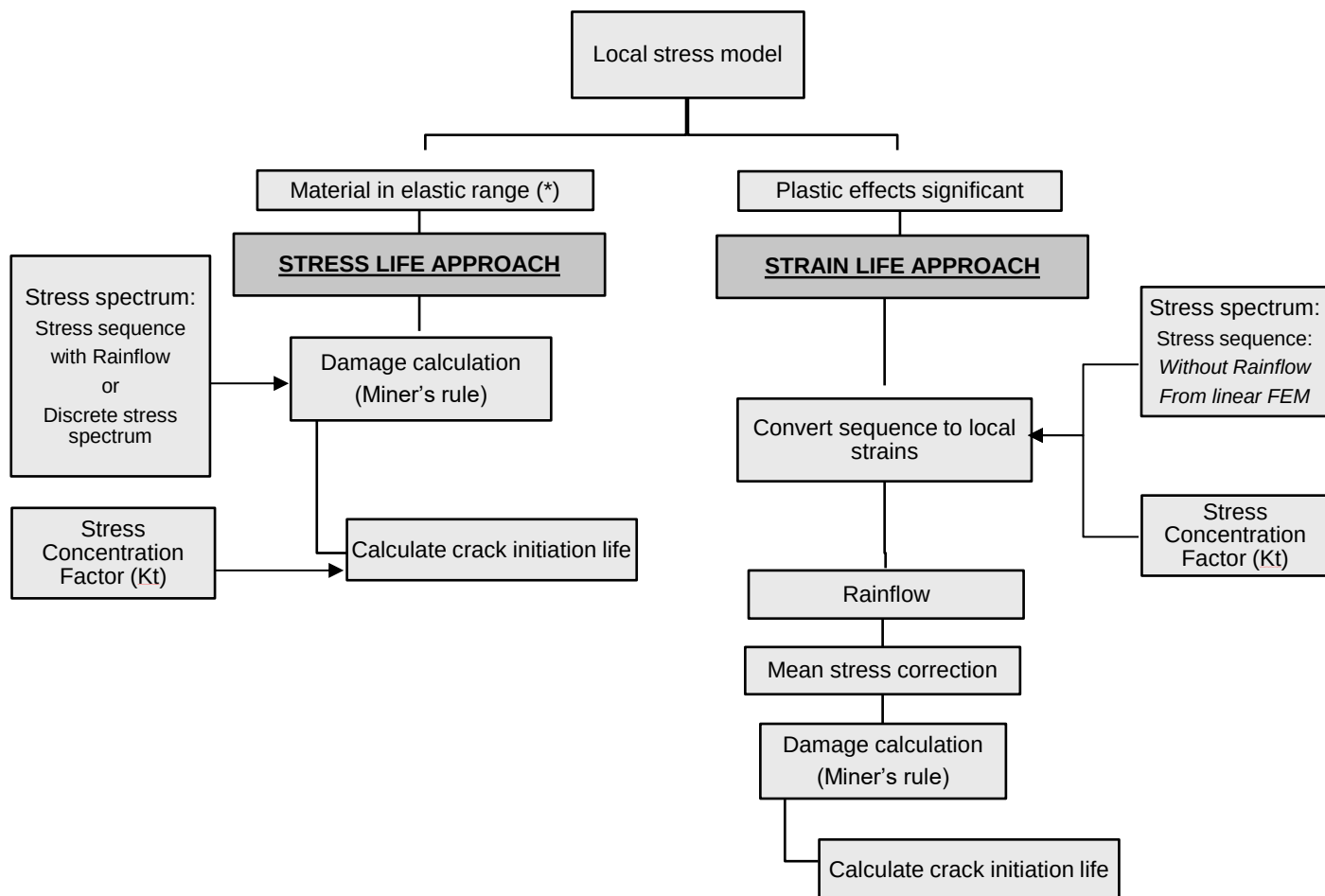


Figure 1. Low cycle fatigue vs High cycle fatigue

The next flowchart shows the steps to be followed in the crack initiation life calculation. Each step is explained in next chapters.

CRACK INITIATION LIFE CALCULATION FOR METALLIC MATERIALS



(*) Peak stresses at the notch area might exceed the yield stress and the stress-life approach be still valid for the analysis. Discuss the right approach with the stress authority assigned to the project.

Figure 2. Crack initiation life calculation flowchart

CRACK INITIATION LIFE CALCULATION FOR METALLIC MATERIALS

2.1 STRESS-LIFE APPROACH

The basis of the stress-life approach is the use of the material stress-life curves (material endurance data) which describe the relation between cyclic stress and the number of cycles to failure.

2.1.1 Stress-Life curves

Material endurance data is represented in the form of Wöhler diagram or Haigh diagram, depending on the material source data:

- Wöhler diagram (S-N curves) in which:

x-axis: Load cycles to failure (N) in log scale.

y-axis: Stress amplitude (σ_a) or maximum stress (σ_{max}) in linear or log scale.

parameter: Mean stress (σ_m) or stress ratio (R)

The upper horizontal asymptote of the Wöhler diagram represent the failure in the 1st cycle, that occurs if $\sigma_{max} = \sigma_u$.

- Haigh diagram, in which:

x-axis: Mean stress (σ_m) in linear scale

y-axis: Stress amplitude (σ_a) in linear scale

parameter: Load cycles to failure (N)

The Haigh diagram is very useful to illustrate the influence of the mean stress on the fatigue strength, whereas the S-N curves are more suitable for the representation of fatigue test results.

Stress-Life curves are obtained from fatigue tests, which are performed by applying a cyclic stress with constant amplitude on specimens until they fail or a large number of cycles is reached. The tests are performed for a number of mean stresses or stress ratios (R) to consider the mean stress effects on the material fatigue properties and for different stress concentration factors (K_t).

The material S-N curves usually represent mean data (50% reliability) but may include an additional factor to produce a higher reliability.

CRACK INITIATION LIFE CALCULATION FOR METALLIC MATERIALS

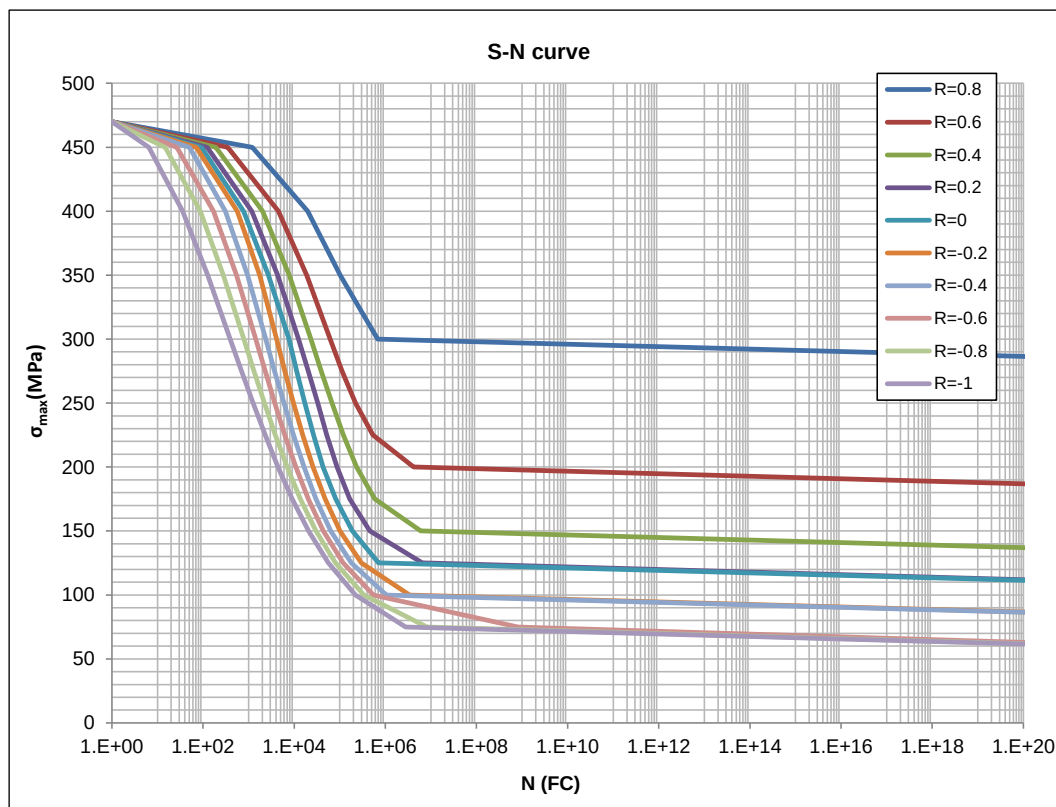


Figure 3. S-N curve ($R = \text{const.}$) for Aluminium 2024-T3, $K_t=2.5$

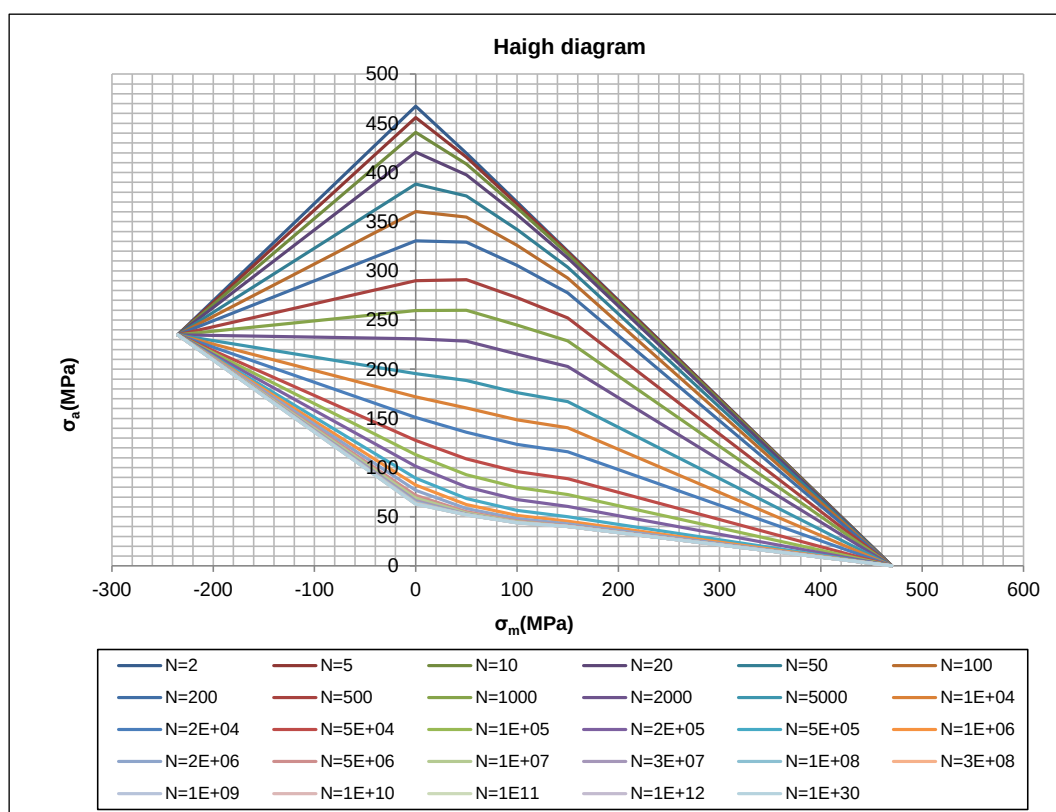


Figure 4. Haigh diagram ($N = \text{const.}$) for Aluminium 2024-T3, $K_t=2.5$

CRACK INITIATION LIFE CALCULATION FOR METALLIC MATERIALS

Interpolation and Extrapolation for different Kt values

S-N data is in general given for only a few Kt values. Very often these are not the Kt values that are required for analysis of a structure. Ref. 5 provides a guideline for the interpolation between and extrapolation beyond Kt values for which S-N data are given.

Note that the calculation method here below exposed is only applicable for high cycle fatigue ($N > 5 \cdot 10^3$).

- $K_{t,SN,i} < K_t < K_{t,SN,i+1}$

The crack initiation life shall be calculated by linear interpolation between $K_{t,SN,i} < K_t < K_{t,SN,i+1}$

$$\log(N) = \log(N_{i+1}) + (\log(N_{i+1}) - \log(N_i)) \cdot \frac{\log(K_t) - \log(K_{t,SN,i+1})}{\log(K_{t,SN,i+1}) - \log(K_{t,SN,i})}$$

Where N_{i+1} and N_i are the number of cycles to failure calculated using the set of S-N data for $K_{t,SN,i+1}$ and $K_{t,SN,i}$ respectively.

- $K_t < K_{t,SN,1}$

Conservatively, the crack initiation life shall be calculated using the S-N data for $K_t = K_{t,SN,1}$

Application of a factor on load would lead to un-conservative results. Therefore, it should not be done.

- $K_t > K_{t,SN,n}$

Conservatively, a factor on load, $f = K_t/K_{t,SN,n}$ shall be used. Then, the crack initiation life shall be calculated using the S-N data for $K_t = K_{t,SN,n}$

2.1.2 Fatigue damage calculation

The most used law for the fatigue damage calculation is the so called "Palmgren-Miner". Miner's rule supposes that the damage suffered by a loaded element at a stress level of σ_i accumulates linearly with the number of cycles of that stresses level. So, the damage caused by n_i cycles will be:

$$d_i = \frac{n_i}{N_i}$$

Where:

- n_i is the number of cycles of the stress spectrum with the same stress levels, (σ_{max-i} , σ_{min-i}). The value of n_i is extracted from the stress spectrum.
- N_i is the number of cycles to failure at the stress levels σ_{max-i} and σ_{min-i} . The value of N is calculated from the corresponding S-N curves or Haigh diagram (see section 2.1.1).

Since the stress spectrum is always defined with two parameters (usually σ_{max} - σ_{min}), it would be necessary a transformation in order to have the correct input parameters to calculate the fatigue life if there is a mismatch between the S-N curve and the spectrum parameters.

CRACK INITIATION LIFE CALCULATION FOR METALLIC MATERIALS

The Palmgren-Miner's cumulative damage law establishes that the total damage for a variable amplitude load spectrum is the sum of the damages of each cycle:

$$D = \sum d_i = \sum \frac{n_i}{N_i}$$

It is assumed that the element failure occurs when the total fatigue damage value exceeds a critical limit $D = 1$:

$$D = \sum \frac{n_i}{N_i} = 1$$

However, when substantiated by test evidences or fleet experience, other values different from the unity could be used.

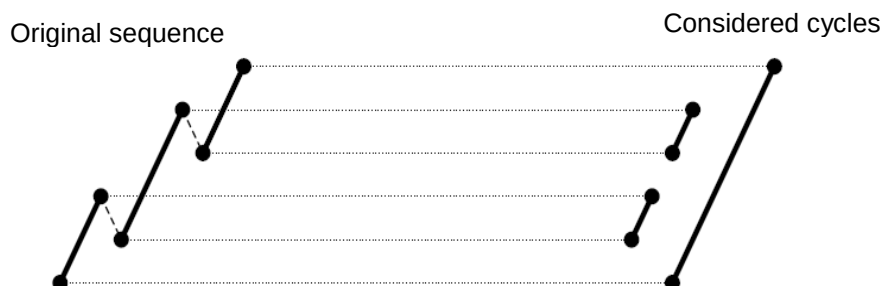
The crack initiation analysis can be performed using either discrete stress spectrum or random stress sequence. The differences in the fatigue damage calculation are explained below:

Random stress sequence:

The stress sequences, also called cycle by cycle spectra, are generated as a randomly distributed cycles among the flights that form the sequence. That means that the cycle's distribution keeps the chronological order of the segments within each flight.

The main limitation of the Palmgren-Miner's rule when calculating the damage is that it does not consider sequence effects, i.e. the order of the loading has no effect in the damage accumulation. However, in some circumstances, cycles of low peak stress followed by high peak stress cause more damage than would be predicted by the rule. In the same manner, it does not consider the effect of an overload of high stress followed by low stress that may have less damage due to the presence of plasticity at notch. Therefore, a cycle counting method is necessary to prevent major errors and enable the reliable application of Miner's rule.

The Rainflow is the counting method used in Airbus D&S. Its basis is the definition of the maximum cycles within a given load or stress sequence (peak-to-peak cycles). The intermediate cycles are determined from the sequence and also considered for the crack initiation life analysis:



Due to the limitations, the Rainflow counting method is applied to each flight independently and considering the stress evolution among the flight. For more information about counting methods, see Ref. 3. Once the sequence has been modified by considering a counting method, the total damage is calculated by summing the damages of each cycle of the stress sequence as:

CRACK INITIATION LIFE CALCULATION FOR METALLIC MATERIALS

$$D = \sum d_i = \sum \frac{1}{N_i}$$

Figure 5 shows the damage calculation for a random stress sequence:

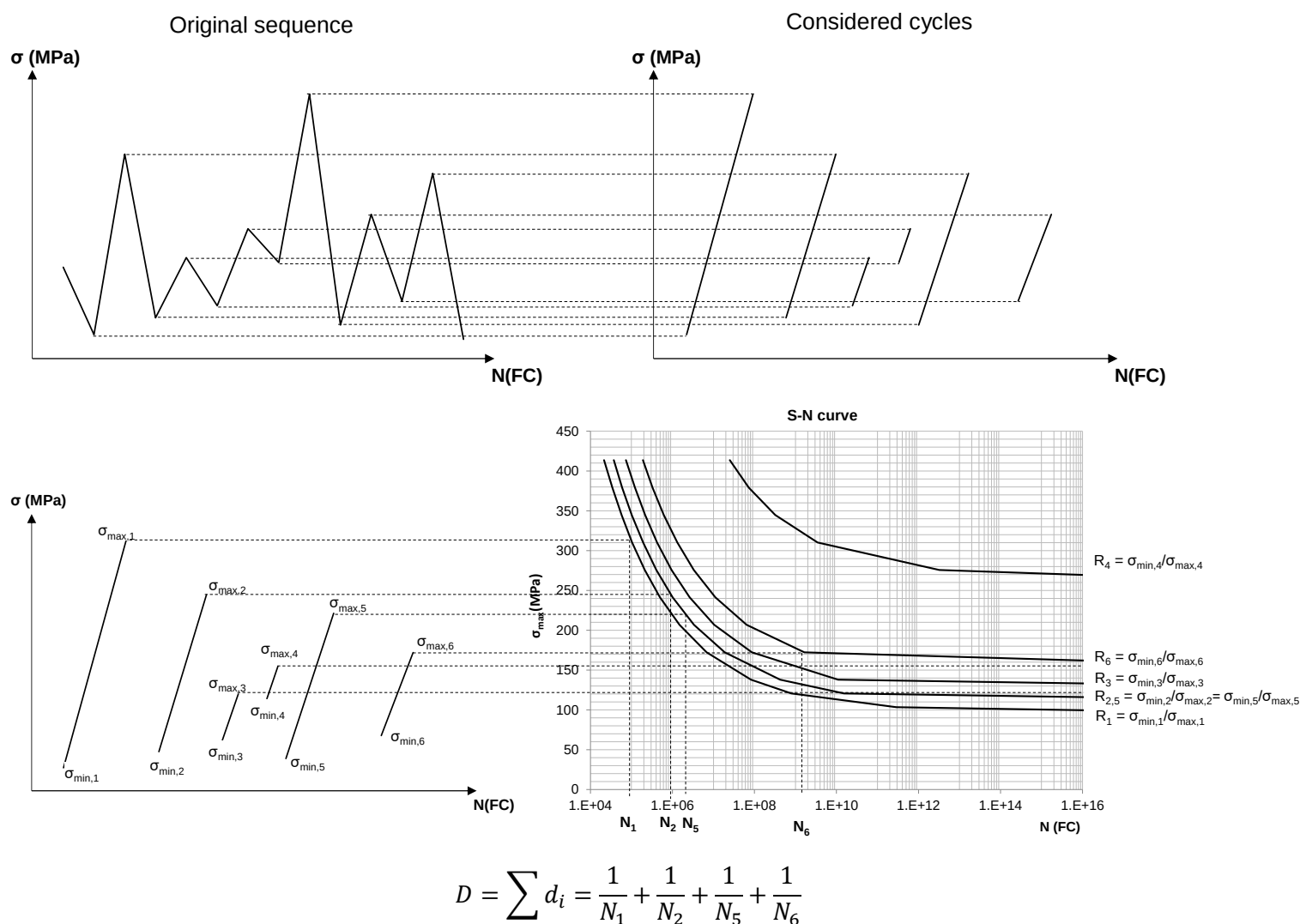


Figure 5. Fatigue damage calculation for random stress sequence

(*) Note that in this example, stress cycles 3 and 4 do not accumulate damage.

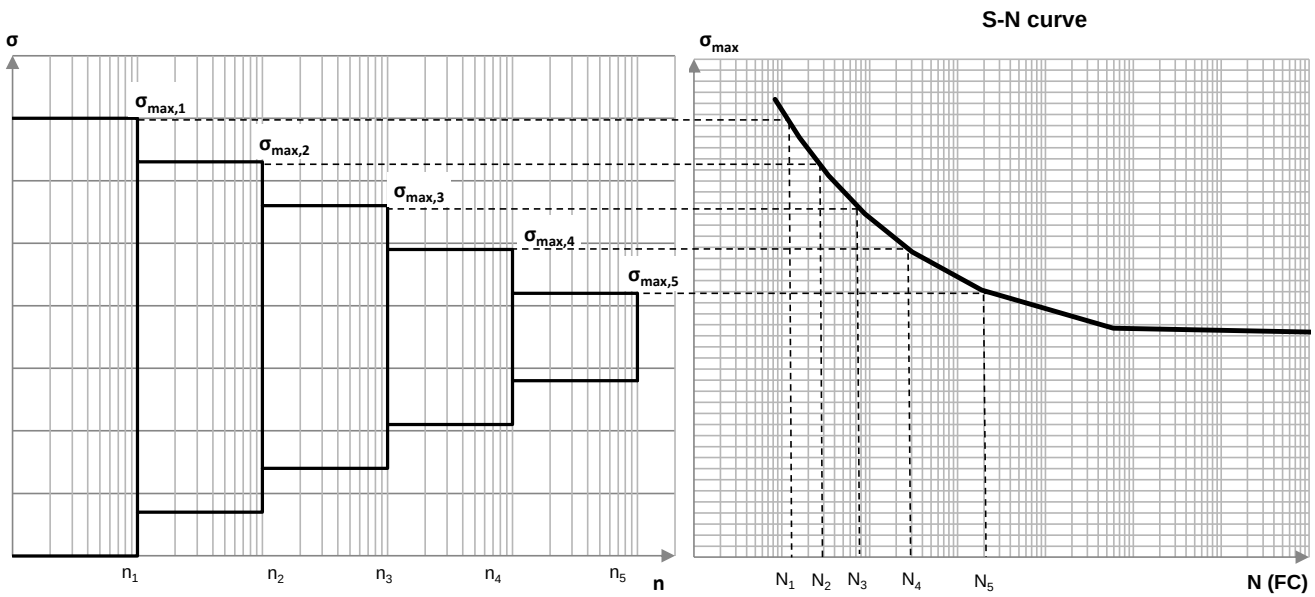
CRACK INITIATION LIFE CALCULATION FOR METALLIC MATERIALS

Discrete stress spectrum

The discrete stress spectrum is a block spectrum defined as a suite of cycle packages, where each package is formed by two parameters that define the cycle ($\sigma_{\max}$ and $\sigma_{\min}$) and the number of times that the cycle repeats itself within the DSG.

The total damage is then calculated by summing the damages of each block of the spectrum (see Figure 6).

$$D = \sum d_i = \sum \frac{n_i}{N_i}$$



$$D = \sum d_i = \frac{1}{N_1} + \frac{1}{N_2} + \frac{1}{N_3} + \frac{1}{N_4} + \frac{1}{N_5} + \frac{1}{N_6}$$

Figure 6. Fatigue damage calculation for discrete stress spectrum

The discrete stress spectrum is obtained directly from the basis spectra without applying any counting method. This means that the ground-air-ground cycles (GAG) could be not included in the spectrum and therefore, not considered in the fatigue damage calculation. This fact could lead to un-conservative results in the fatigue damage calculation. To prevent it, the GAG cycles shall be calculated from the maximum and minimum stress of each flight for each flight mission and considered in the damage calculation as follows:

$$D = \sum_{m=1}^{N_{missions}} \left(\sum \frac{n_i}{N_i} + \sum_{GAG} \frac{n}{N} \right)$$

Note that if the block spectrum used for the crack initiation analysis is obtained from a stress sequence for which a Rainflow counting method has been applied, there is no need to consider the damage produced by the GAG cycle.

CRACK INITIATION LIFE CALCULATION FOR METALLIC MATERIALS

2.2 STRAIN-LIFE APPROACH

Most engineering structures are designed such that the nominal stress remains elastic under service loads. For those elements, the stress-life approach is adequate to predict the fatigue damage. However, in some applications the stress concentrations or the high service loads applied cause plastic strains. In these scenarios, the strain-life approach has to be considered because the stress-life approach leads to unrealistic results.

To apply the strain-life approach the fatigue spectra should be expressed in terms of local strains. The fatigue spectra are usually obtained from FEMs in terms of far-field/gross stresses. Therefore, it needs to be converted from nominal/gross stresses to local strains.

There are several methods to obtain the local stress-strain values, like Neuber's rule (Ref. 8), Glinka's (Ref. 9) or Seeger's method (Ref. 10).

The Neuber's and Glinka's approach are suitable for those components in which plasticity is limited to a small area around the stress concentration area. Nevertheless, Glinka's underestimates stresses at the notch for some cases, which can lead to non-conservative predictions, and this method is not recommended.

For those locations where nominal/gross stresses and strains are so high that cause plasticity across the component, Seeger's method proposed an alternative approach that gives more realistic results.

The vast majority of airframe structural components are designed such that nominal/gross stresses remain elastic under service loads. This is why, from the three above mentioned methods, Neuber's rule is the most used in the fatigue field and the one chosen in Airbus D&S to evaluate the local stresses and strains at stress concentration areas. Deviations from this approach have to be accepted by the stress authority of the project. The basis of Neuber's rule is explained in section 2.2.2.

Once the fatigue spectrum is expressed in terms of real strains, the fatigue damage is calculated using the strain-life curves applying the Palmgren-Miner's rule (section 2.2.4).

The material fatigue properties required for the strain-life approach are the cyclic stress-strain curve (see 2.2.1) and the strain-life curve, derived from the Coffin-Manson relation (see 2.2.3).

2.2.1 Cyclic stress-strain curve

The true stress-strain relationship obtained for a single load application is the monotonic curve, also known as the Ramberg-Osgood's equation:

$$\varepsilon = \varepsilon_e + \varepsilon_p = \frac{\sigma}{E} + \left(\frac{\sigma}{K}\right)^{1/n}$$

Where n and K are the strain hardening exponent and the strength coefficient respectively.

An equivalent formulation can be found in some literature, such as Ref. 13, where a plastic strain of 0.002 is assumed at the yield point:

$$K = \frac{F_{ty}}{0.002^n} \Rightarrow \varepsilon = \frac{\sigma}{E} + 0.002 \cdot \left(\frac{\sigma}{F_{ty}}\right)^{1/n}$$

CRACK INITIATION LIFE CALCULATION FOR METALLIC MATERIALS

In a cyclic loaded component the material response is a hysteresis loop of true stress and true strain. Similar to the monotonic relationship, the material cyclic stress-strain curve can be written as:

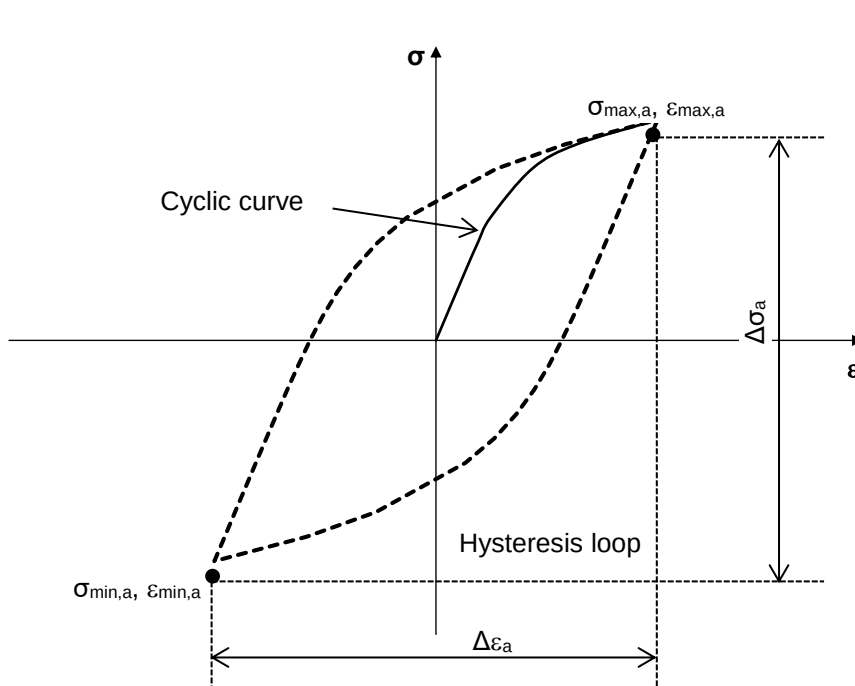
$$\varepsilon = \frac{\sigma}{E} + \left(\frac{\sigma}{K'}\right)^{1/n'}$$

Where σ and ε are the cyclically stable stress and strain amplitudes respectively, and K' and n' are the cyclic strength coefficient and the cyclic strain hardening exponent respectively.

For most materials the hysteresis loop data can be derived by scaling the cyclic stress-strain curve using Massing's equation (see Ref. 4). It allows obtaining the hysteresis loop by doubling the cyclic stress-strain curve as shown in Figure 7.

The hysteresis loop equation, in terms of ranges, is obtained as:

$$\frac{\Delta\varepsilon}{2} = \frac{\Delta\sigma}{2E} + \left(\frac{\Delta\sigma}{2K'}\right)^{1/n'} \Rightarrow \Delta\varepsilon = \frac{\Delta\sigma}{E} + 2\left(\frac{\Delta\sigma}{2K'}\right)^{1/n'}$$



Cyclic curve:

$$\varepsilon_{max,a} = \frac{\sigma_{max,a}}{E} + \left(\frac{\sigma_{max,a}}{K'}\right)^{1/n'}$$

Hysteresis loop:

$$\Delta\varepsilon_a = \frac{\Delta\sigma_a}{E} + 2\left(\frac{\sigma_a}{2K'}\right)^{1/n'}$$

Figure 7. Hysteresis loop and cyclic curve

CRACK INITIATION LIFE CALCULATION FOR METALLIC MATERIALS

During the cyclic loading, a material can have different transient behavior, depending on the initial conditions (i.e., heat treatment):

- Cyclically strain-harden
- Cyclically strain-soften
- Cyclically stable
- Mixed behavior

To estimate if a material will harden or soften, it is useful to check the ratio of ultimate strength (σ_{ult}) to the yield strength (σ_{yield}) (see Ref. 6):

- $\frac{\sigma_{ult}}{\sigma_{yield}} > 1.4$ The material cyclically harden
- $\frac{\sigma_{ult}}{\sigma_{yield}} < 1.2$ The material cyclically soften
- $1.2 \leq \frac{\sigma_{ult}}{\sigma_{yield}} \leq 1.4$ The material can soften or harden, but it is not expected to suffer big changes in the cyclic strain-stress response.

Another parameter to estimate the behaviour of the material is the monotonic strain-hardening exponent:

- $n > 0.2$ The material cyclically harden
- $n < 0.2$ The material cyclically soften

This behaviour is expected to occur in the early crack initiation life. After this transient softening or hardening, the material reaches a cyclically stable condition. It is considered that this stable state is reached between 20 and 40% of the total life. Thus, fatigue properties are given at the 50% of the crack initiation life and the hardening or softening of the material is not considered in the crack initiation life calculation when using Airbus D&S tools.

CRACK INITIATION LIFE CALCULATION FOR METALLIC MATERIALS

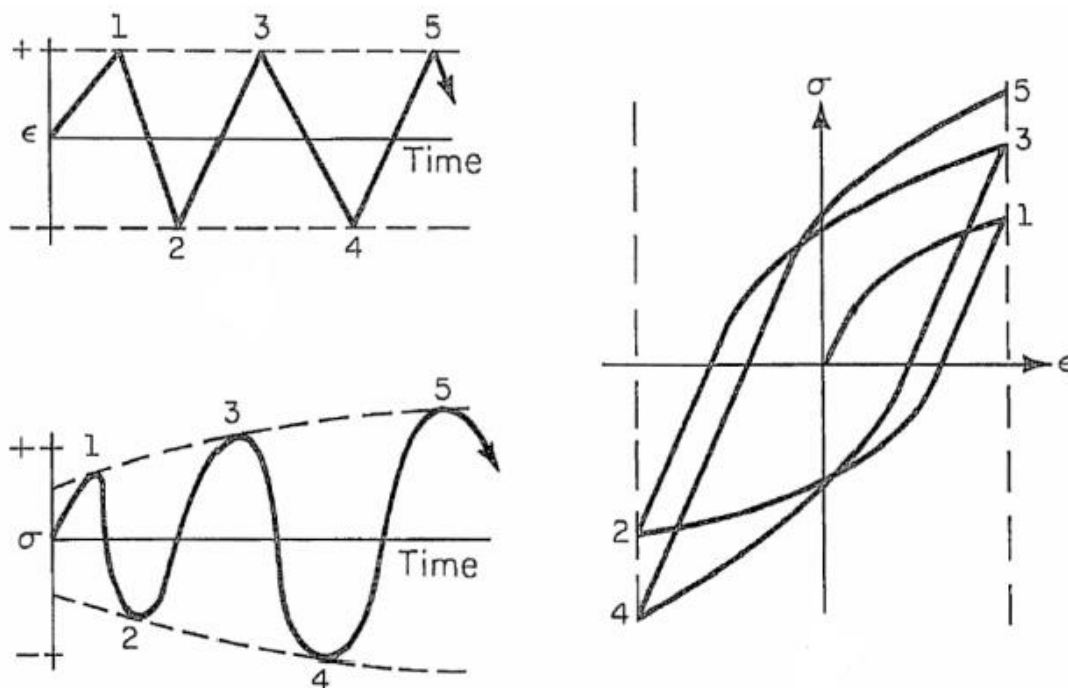


Figure 8. Cyclic strain hardening. Under a constant strain amplitude, the stress level is increased (from Ref. 7)

CRACK INITIATION LIFE CALCULATION FOR METALLIC MATERIALS

2.2.2 Neuber's rule

Neuber's method states that the product of stress and strain in a concentration area is a constant that is independent of the material stress-strain law. That means that the local elastic-plastic stress-strain product is equal to the linearly calculated stress-strain product:

$$\sigma_r \cdot \varepsilon_r = \sigma_e \cdot \varepsilon_e$$

Where ε_r and σ_r are the true local elastic-plastic strain and stress and ε_e and σ_e are the strain and stress which would be calculated if the material was infinitely elastic.

Neuber's rule can be expressed in terms of net nominal stress as:

$$\sigma_r \cdot \varepsilon_r = \frac{(K_t \cdot \sigma)^2}{E}$$

Neuber's rule can be represented by a hyperbola in the stress-strain diagram; this hyperbola is defined by the notch stress calculated for a stress concentration area using the material linear elastic stress-strain law. The intersection of the Neuber's hyperbola with the material elastic-plastic stress-strain law defines the actual elastic-plastic strain according to the Neuber's method (see Ref. 1 and Figure 9).

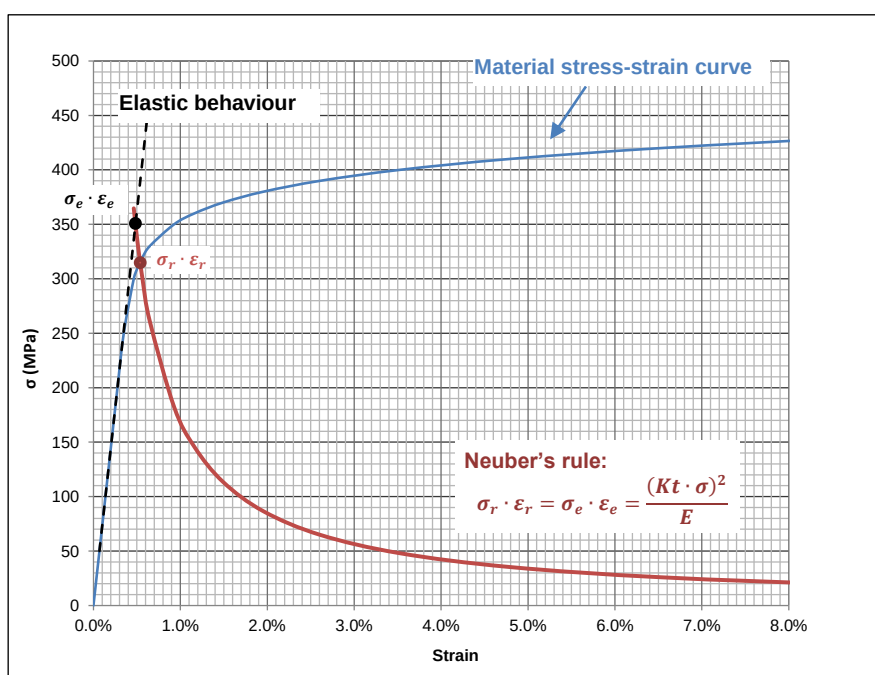


Figure 9. Neuber's rule graphical solution

CRACK INITIATION LIFE CALCULATION FOR METALLIC MATERIALS

2.2.3 Strain-Life curves

The strain-life approach is based on the use of the material strain life curve. The strain life curve is obtained by cyclic strain-controlled tests where the total strain amplitude (ϵ_a) is obtained as the sum of the elastic (ϵ_{ae}) and the plastic (ϵ_{ap}) strains measurements. From Basquin, Coffin and Manson expressions, the strain-life relationship can be represented as:

$$\frac{\Delta\epsilon}{2} = \frac{\sigma'_f}{E} (2N_f)^b + \epsilon'_f (2N_f)^c$$

The constants in this equation are:

- b is the fatigue strength exponent (Basquin's exponent)
- σ'_f is the fatigue strength coefficient
- c is the fatigue ductility exponent (the Coffin-Manson exponent)
- ϵ'_f is the fatigue ductility coefficient, which is the plastic strain amplitude at $2N_f = 1$
- E is the elastic modulus
- $2N_f$ are the reversals to failure

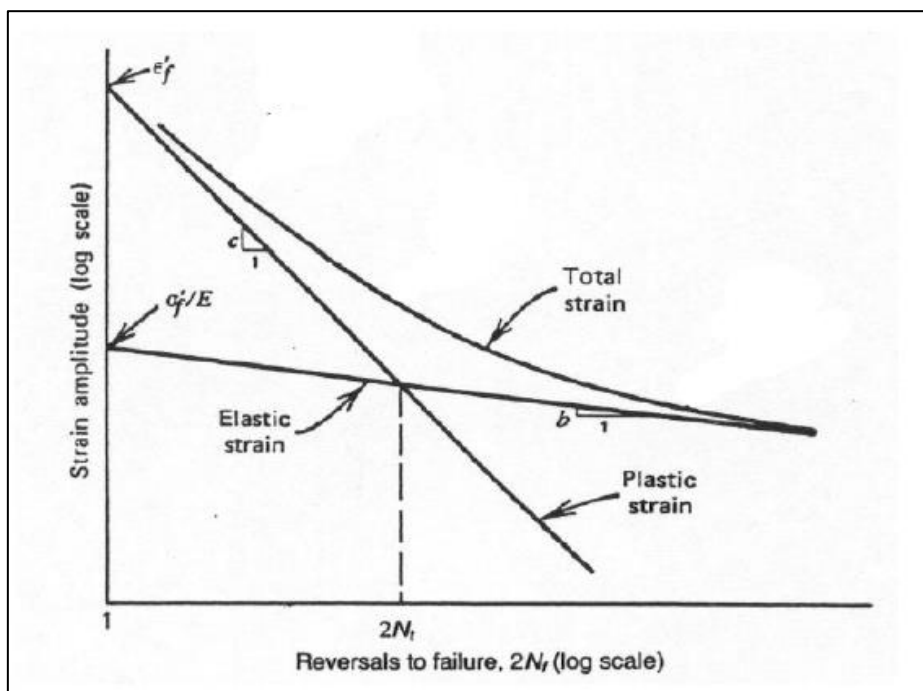


Figure 10. Strain-Life curve (from Ref. 7)

In these curves, the fatigue data of the notch area of a component is only represented under full reversed stress ($\sigma_m = 0$, $R = -1$). This is the reason why a mean stress correction is needed when using this approach.

CRACK INITIATION LIFE CALCULATION FOR METALLIC MATERIALS

The material ε -N curves usually represent mean data (50% reliability), but they may include an additional factor to produce higher reliability.

2.2.4 Fatigue damage calculation

The stepwise ε -N procedure for calculating the fatigue damage by applying Neuber's rule can be summarized as follows (see Ref. 2):

1. Local stress and strain calculation.
2. Cycle counting
3. Mean stress correction
4. Damage calculation

A description of each step follows:

Local stress and strain calculation

First of all, the input stress sequence is transformed from linear net nominal stress to real local strains. The most common methodology used to relate the local and the nominal stress and strain is the Neuber's rule (see chapter 2.2.2).

The real strain sequence is calculated following the linear stress sequence, converting each stress state in real strain state taking into account the material plasticity and the material hysteresis curve. When following the stress sequence, next criteria shall be considered:

- If the current peak or valley is higher, in absolute value, than all the previous valleys and peaks, the real stress (σ_r) and strain (ε_r) are calculated as:

$$\sigma_r \cdot \varepsilon_r = \frac{(Kt \cdot \sigma(i))^2}{E}$$

$$\sigma_r(i) = F_h(\sigma_r, \varepsilon_r)$$

$$\varepsilon_r = \frac{\sigma_r \cdot \varepsilon_r}{\sigma_r(i)}$$

Where F_h is the function defined by the hysteresis curve (see 2.2.1.)

- Descending:
 - If the current valley is lower or equal than the previous lowest valley, the origin is the previous highest peak.

$$\sigma(i) \leq \sigma_{min} \Rightarrow \Delta\sigma = \sigma_{max} - \sigma(i)$$

$$\sigma_{r0} = \sigma_{rmax}$$

$$\varepsilon_{r0} = \varepsilon_{rmax}$$

CRACK INITIATION LIFE CALCULATION FOR METALLIC MATERIALS

- If the current valley is higher than the previous lowest valley, the origin is the current peak.

$$\sigma(i) > \sigma_{min} \Rightarrow \Delta\sigma = \sigma(i-1) - \sigma(i)$$

$$\sigma_{r0} = \sigma_r(i-1)$$

$$\varepsilon_{r0} = \varepsilon_r(i-1)$$

The real stress and strain increments are calculated by applying Neuber's rule:

$$\Delta\sigma_r \cdot \Delta\varepsilon_r = \frac{(Kt \cdot \Delta\sigma)^2}{E}$$

$$\Delta\sigma_r = F_h(\Delta\sigma_r, \Delta\varepsilon_r)$$

$$\Delta\varepsilon_r = \frac{\Delta\sigma_r \cdot \Delta\varepsilon_r}{\Delta\sigma_r}$$

Where F_h is the function defined by the hysteresis curve (see 2.2.1.).

Finally, the real stress and strain values for the valley are calculated as:

$$\sigma_r(i) = \sigma_{r0} - \Delta\sigma_r$$

$$\varepsilon_r(i) = \varepsilon_{r0} - \Delta\varepsilon_r$$

If the current valley is lower than all the previous ones, the value of the minimum valley is updated:

$$\sigma_{min} = \sigma(i)$$

$$\sigma_{r-min} = \sigma_r(i)$$

$$\varepsilon_{r-min} = \varepsilon_r(i)$$

- Ascending:

- If the current peak is higher or equal than the previous highest peak, the origin is the previous lowest valley.

$$\sigma(i) \geq \sigma_{max} \Rightarrow \Delta\sigma = \sigma(i) - \sigma_{min}$$

$$\sigma_{r0} = \sigma_{rmin}$$

$$\varepsilon_{r0} = \varepsilon_{rmin}$$

- If the current peak is lower than the previous highest peak, the origin is the current valley.

$$\sigma(i) < \sigma_{min} \Rightarrow \Delta\sigma = \sigma(i-1) - \sigma(i)$$

$$\sigma_{r0} = \sigma_r(i-1)$$

$$\varepsilon_{r0} = \varepsilon_r(i-1)$$

CRACK INITIATION LIFE CALCULATION FOR METALLIC MATERIALS

The real stress and strain increments are calculated by applying Neuber rule:

$$\Delta\sigma_r \cdot \Delta\varepsilon_r = \frac{(Kt \cdot \Delta\sigma)^2}{E}$$

$$\Delta\sigma_r = F_h(\Delta\sigma_r, \Delta\varepsilon_r)$$

$$\Delta\varepsilon_r = \frac{\Delta\sigma_r \cdot \Delta\varepsilon_r}{\Delta\sigma_r}$$

Where F_h is the function defined by the hysteresis curve (see 2.2.1.).

Finally, the real stress and strain values for the peak are calculated as:

$$\sigma_r(i) = \sigma_{r0} + \Delta\sigma_r$$

$$\varepsilon_r(i) = \varepsilon_{r0} + \Delta\varepsilon_r$$

If the current peak is higher than all the previous ones, the value of the maximum peak is updated:

$$\sigma_{max} = \sigma(i)$$

$$\sigma_{r-max} = \sigma_r(i)$$

$$\varepsilon_{r-max} = \varepsilon_r(i)$$

Cycle counting

Once the input sequence has been transformed from linear net nominal stress to real local strains, the “Full-Rainflow” is applied in order to consider the sequence effects (see Ref. 3 for more information).

Mean stress correction

Cyclic fatigue properties of materials are obtained only from completely reversed stress ($R = -1$, $\sigma_m = 0$) and constant amplitude strain-controlled tests (see section 2.2.1).

Mean stresses different from zero have an important influence on the crack initiation life of the component. It can either increase the crack initiation life with a compressive mean stress or decrease it with a tensile value. Since the material fatigue properties are only available for zero mean stress, a mean stress correction is needed to transform the original strain sequence to a new one, with the same fatigue damage but where every cycle has zero mean stress.

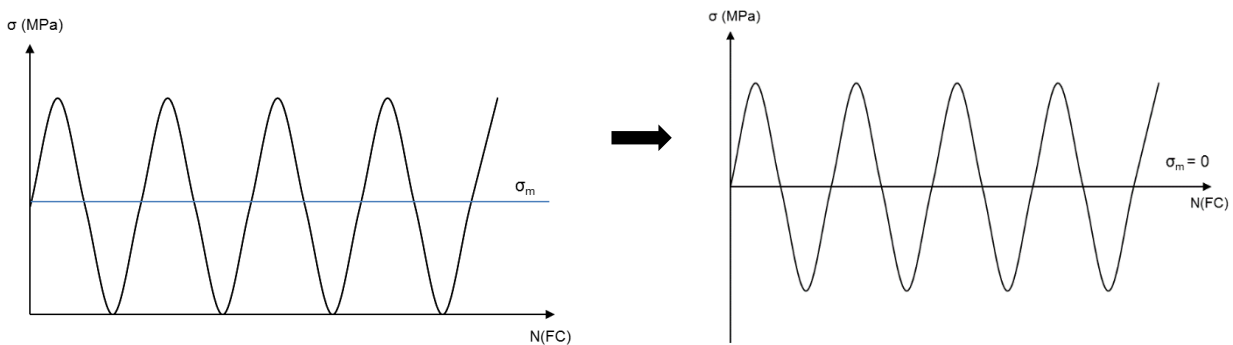


Figure 11. Mean stress correction

CRACK INITIATION LIFE CALCULATION FOR METALLIC MATERIALS

There are different approaches to correct the mean stress effect such as the ones proposed by Morrow, Goodman, Gerber, Soderberg or Smith-Watson-Topper:

- The Smith-Watson-Topper:

$$\left(\frac{\Delta\varepsilon}{2}\right)_{EQ} = \sqrt{\frac{\sigma_{max}}{E} \cdot \left(\frac{\varepsilon_{max} - \varepsilon_{min}}{2}\right)}$$

Where σ_{max} is the maximum stress of the current cycle, and ε_{max} and ε_{min} are the maximum and minimum strain of the current cycle.

This correction only works in the tensile region. Damage of compression cycles is considered null.

- Modified Goodman:

$$\left(\frac{\Delta\varepsilon}{2}\right)_{EQ} = \frac{\frac{\Delta\varepsilon}{2}}{\left(1 - \frac{\sigma_m}{\sigma_{ult}}\right)}$$

This option is recommended for high-strength/low ductility materials in Ref. 11.

- Gerber:

$$\left(\frac{\Delta\varepsilon}{2}\right)_{EQ} = \frac{\frac{\Delta\varepsilon}{2}}{\left(1 - \left(\frac{\sigma_m}{\sigma_{ult}}\right)^2\right)}$$

This option is recommended for ductile materials in Ref. 11.

- Soderberg:

$$\left(\frac{\Delta\varepsilon}{2}\right)_{EQ} = \frac{\frac{\Delta\varepsilon}{2}}{\left(1 - \frac{\sigma_m}{\sigma_{yield}}\right)}$$

From the different approaches exposed above, the one proposed by Smith-Watson-Topper (Ref. 12) is the most one used because it provides reasonable results in most cases and it does not need additional material data.

Damage calculation

The damage of each closed hysteresis loop is finally calculated as:

$$d_i = \frac{1}{N_i}$$

Where N_i is the number of cycles to failure for the cycle i . The value of N is calculated from the material ε - N curves for $R = -1$ (see 2.2.3).

**CRACK INITIATION LIFE CALCULATION FOR
METALLIC MATERIALS**

The total damage is calculated by summing the damages of each cycle of the sequence (Miner's rule):

$$D = \sum d_i = \sum \frac{1}{N_i}$$

CRACK INITIATION LIFE CALCULATION FOR METALLIC MATERIALS

3 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
b	---	Fatigue strength exponent (Basquin's exponent)
c	---	Fatigue ductility exponent (Coffin-Manson exponent)
D	---	Damage
d _i	---	Damage of each cycle i of the spectrum
DSG	FC	Design Service Goal
E	MPa	Young modulus
F _h	---	Hysteresis curve
FL	FC	Crack initiation life
GAG	---	Ground-Air-Ground
K	---	Strength coefficient
K'	---	Cyclic strength coefficient
K _t	---	Stress concentration factor
n	---	Strain hardening exponent
n'	---	Cyclic strain hardening exponent
n _i	FC	Number of cycles of the stress spectrum with the same stress levels
N, N _f	FC	Number of cycles to failure
N _{sp}	FC	Number of flights for which the stress spectrum has been generated
R	---	Stress Ratio = $\sigma_{\min} / \sigma_{\max}$
ε	---	Strain
ε_e	---	Elastic strain
ε'_f	---	Fatigue ductility coefficient
$\varepsilon_{\max}$	---	Maximum strain
$\varepsilon_{\min}$	---	Minimum strain
ε_r	---	Elastic-plastic strain
σ'_f	---	Fatigue strength coefficient
σ_a	MPa	Stress amplitude = $(\sigma_{\max} - \sigma_{\min}) / 2$
σ_e	MPa	Elastic stress
σ_m	MPa	Mean stress = $(\sigma_{\max} + \sigma_{\min}) / 2$
$\sigma_{\max}$	MPa	Maximum stress
$\sigma_{\min}$	MPa	Minimum stress
σ_r	MPa	Elastic-plastic stress
σ_{ult}	MPa	Ultimate stress
σ_{yield}	MPa	Yield stress

CRACK INITIATION LIFE CALCULATION FOR METALLIC MATERIALS

REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	HSB 34201-01	A	Interpretation of local stress concentrations from linear Finite Element analysis using the Neuber method.
Ref. 2	NT-T-ADP-08018	A	Manual teórico del programa FATIGA Versión 7
Ref. 3	NT-T-ADP-02015	A	Manual teórico del programa SECUENCIA
Ref. 4	G.Massing	1926	Eigenspannungen und Verfestigung beim Messing
Ref. 5	HSB 62131	A	Guidelines for the prediction of the fatigue life for Kt values different to available databases
Ref. 6	ESDU 04022		An introduction to low-cycle fatigue phenomena.
Ref. 7	Bannantine,J., Comer,J., Handrock,J.		Fundamentals of metal fatigue analysis
Ref. 8	Neuber H.	1961	Theory of stress concentration for shear strained bodies with arbitrary stress-strain law.
Ref. 9	Molski K, Glinka G.	1981	A method of elastic-plastic stress and strain calculation at a notch root
Ref. 10	T.Seeger	1980	Generalized application of Neuber's rule.
Ref. 11	Schijve J.	2009	Fatigue of structures and materials
Ref. 12	K.N.Smith,P.Watson, T.H. Topper	1970	A Stress-Strain function for the fatigue of metals
Ref. 13	MMPDS	13	Metallic Materials Properties Development and Standardization

CRACK INITIATION LIFE CALCULATION FOR METALLIC MATERIALS

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>

Table 1. *Software programs referenced in this chapter*

RECORD OF REVISIONS

<i>Revision Date</i>	<i>Authors</i>	<i>Change Reason</i>	<i>Pages/Sub-chapters affected</i>
1.0 06/03/2020	M.Carmen Blanquer Darío Fernández	First issue	All pages
1.1 18/01/2021	M.Carmen Blanquer Darío Fernández	Minor and wording changes along de document.	---

Table 2. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

<i>Dpt. \ Site</i>	<i>Getafe</i>	<i>Manching</i>
Stress Analysis (TEPAA)	Javier Gómez-Escalonilla 18/01/2021	Holger Hickethier 18/01/2021
Stress Methods (TEPAP-TL4)	Darío R. Fernández Villalba 18/01/2021	-

Table 3. *Approval signatures table for last revision of this chapter*